

Project Proposal – Explainability Applied to RNNs

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In today's world, Recurrent Neural Networks (RNNs) have become one of the most popular architectures in artificial intelligence, especially for sequential data analysis applications such as time-series prediction and natural language processing. Unfortunately, the black-box nature of AI models is frequently exacerbated by the inherent complexity of RNNs, making them harder to analyse and comprehend. This opacity presents serious difficulties, particularly in industries like finance where risk management and regulatory compliance depend heavily on transparency. Therefore, even in RNN networks, implementing eXplainable Artificial Intelligence (XAI) techniques becomes essential. Through the improvement of RNN-based model interpretability, XAI techniques facilitate compliance with regulations, build confidence, and unleash the full potential of RNNs in a range of real-world applications by helping stakeholders comprehend the underlying decision-making processes.

In this project, we aim to explore the different explainability methods currently available for Recurrent Neural Networks. We will first explore the available literature, where we found the following relevant papers. We will split them into topics:

- **Feature importance, relationships, and effects**
 - **Feature importance** What Is Feature Importance in Machine Learning?
<https://www.baeldung.com/cs/ml-feature-importance>
 - **Variable effects (linear or nonlinear)**
Response Prediction for Linear and Nonlinear Structures Based on Data-Driven Deep Learning <https://www.mdpi.com/2076-3417/13/10/5918>
 - **Interaction effects:** Correlation within input data such as: Temporal, Spatial, Cyclic, Long-range, Cross, Contextual or Hierarchical.
<https://aclanthology.org/N16-1174.pdf> arxiv.org/pdf/1704.06857.pdf
[1409.0473.pdf \(arxiv.org\)](https://arxiv.org/pdf/1409.0473.pdf) <https://aclanthology.org/W18-5408.pdf> [LSTM](#)
- **Type of methods**
 - **Agnostic methods:** These methods work on every type of model, i.e., they are not limited to NN/RNN/LSTM/etc. We are currently considering the following techniques (however not limiting ourselves to these and these only), and found several sources, some of which are indicated below.
 - **SHAP** <https://arxiv.org/abs/1705.07874> 05/2017
 - **PDP, ICE, ALE** <https://neptune.ai/blog/explainability-auditability-ml-definitions-techniques-tools> <https://hastie.su.domains/ElemStatLearn/>
<https://arxiv.org/abs/1309.6392> <https://arxiv.org/abs/1612.08468>
 - **Permutation importance**
 - **LIME** <https://arxiv.org/abs/1602.04938>
 - **Specific methods:** These models work on specific model architectures. We found some basic sources but are considering including newer techniques.
 - **Integrated gradients** <https://arxiv.org/abs/1703.01365>
 - **Saliency Maps** <https://arxiv.org/abs/1911.11293>
 - **XRAI** <https://arxiv.org/abs/1906.02825> 3 main contributions

Those articles explain how each method we have found works and in which cases it is usually applied. Also, they explain a little bit how to implement them.

- **Explainability in RNNs for NLP**

- Model Explainability in Deep Learning Based Natural Language Processing

<https://arxiv.org/pdf/2106.07410.pdf>

This article dives into explaining NLP machine learning models. It compares methods for explanation, focusing on NLP tasks. One technique, Layer-wise Relevance Propagation, helps pinpoint which words matter most in a specific NLP prediction. This local explainability can be used to understand the model's overall variable importance and identify weaknesses, making NLP models less like black boxes.

- **Benchmarking**

- Towards Benchmarking Explainable Artificial Intelligence Methods: <https://arxiv.org/abs/2208.12120>
- Precise Benchmarking of Explainable AI Attribution Methods <https://arxiv.org/abs/2308.03161>

Benchmarking for explainability techniques seems to be a field that hasn't been researched much. At first glance, there doesn't seem to be specific XAI benchmarks for RNNs. Also, some of them are based on human benchmarking, which is not ideal. In this project we have idea to explore and maybe develop some new benchmarking methods.

There are 3 main milestones for this project: Project progress report 1 12th April; Project progress report 2 19th April; Paper + slides + code + team declaration 22nd April. By 12th April, we aim to have implemented the agnostic XAI methods and be able to apply them to different RNNs. Since the time is tight, ideally, we will have implemented some specific methods as well by this date. By the 19th of April, we aim to have finished implementing all XAI methods and started on benchmarking them, so that we can spend the last couple of days after that finishing the benchmarking and preparing the final paper.

For this project, we aim to explore the different explainability methods for Recurrent Neural Networks (RNN). We will separate them into 2 different categories. First, we will explore agnostic explainable Artificial Intelligence (XAI) algorithms that can be used for any model and apply them to the explanation of RNN. Secondly, we will implement specific methods for RNN, methods that are specifically built for RNN architectures. Finally, we will explore the different benchmarking and evaluation methods that exist for explainability and evaluate the different models based on these benchmarks. Since there isn't much work done on benchmarking explainability methods for RNN, we would like to explore and propose new benchmarking methods.

As for the data, we will first use RNN trained to analyze numerical data, for example time series forecasting. Our intention by using simple numerical data is that it seems to be simpler to implement explainability methods and to make sense of them. After all methods we want to analyze work with these networks, we will try some XAI methods with RNN trained for natural language processing tasks, namely text classification.